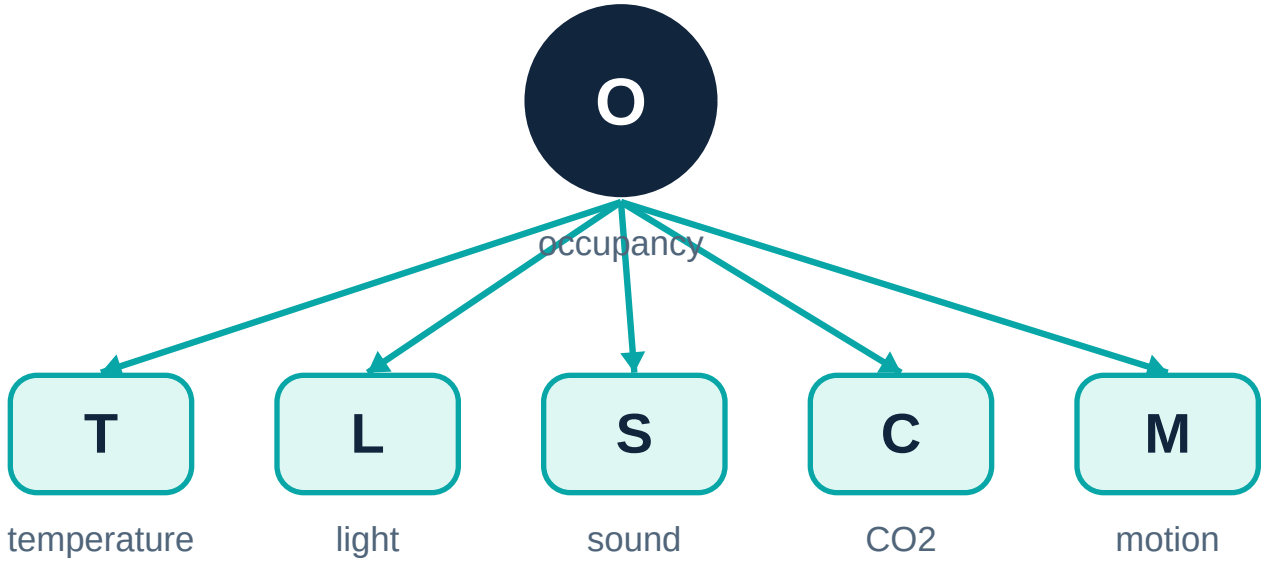
# MODEL AND ASSUMPTION

### Naive Bayes structure

Occupancy is the root; each sensor receives one directed edge.




## Variable meanings

**O** = occupancy

**T** = temperature sensor reading

**L** = light sensor reading

**S** = sound sensor reading

**C** = CO2 sensor reading

**M** = motion sensor reading

**MODELLING ASSUMPTION**

Sensor features are conditionally independent of each other given occupancy (naive Bayes structure).

# PROTOCOL AND EVIDENCE

**Fixed chronological split:** train before test in time. Train: 7,090 rows, 2017-12-22T10:49:41 to 2017-12-26T00:35:20. Test: 3,039 rows, 2017-12-26T00:35:51 to 2018-01-11T09:00:09.

| Metric | prior only | full sensors | without C |
|---|---|---|---|
| n | 3039.0 | 3039.0 | 3039.0 |
| brier | 0.10428 | 0.02350 | 0.02630 |
| mean_loss | 0.90326 | 0.08292 | 0.08424 |
| TN | 0.0 | 2743.0 | 2744.0 |
| FP | 2745.0 | 2.0 | 1.0 |
| FN | 0.0 | 50.0 | 51.0 |
| TP | 294.0 | 244.0 | 243.0 |

**Numerical result:** Brier score is 0.02350 with full sensors versus 0.02630 without C — a difference of 0.00280.

# ENGINEERING CONCLUSION

## Single incident from cell 19

| Condition | Probability | Decision |
|---|---|---|
| full sensors | 1.297417e-07 | False |
| without C | 8.311271e-05 | False |

**Decision did not flip:** empty in both conditions.

## Important limitation

The occupancy balance differs between periods: train class counts are **0: 5483, 1: 1607**; test counts are **0: 2745, 1: 294**. Results may not generalize to a differently balanced deployment period.

## Further validation

Repeat the outage experiment across multiple sensors and on a re-shuffled or cross-validated split, rather than relying on one fixed chronological split.